Coursework Assignment: Full Custom Design of a Digital to Analogue Converter (DAC) in 0.18μm CMOS

Department of Electrical and Electronic Engineering

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1 Introduction

For this coursework, you are required to design an integrated circuit in a 180nm generic CMOS process using the Cadence Virtuoso tool suite. A typical design flow for full custom ICs is shown in Fig.1, including schematic design, circuit simulation, layout implementation, packaging and testing. This project will focus on the first three stages only (i.e. up to and including the layout) for the design of a digital to analogue converter (DAC) [1].

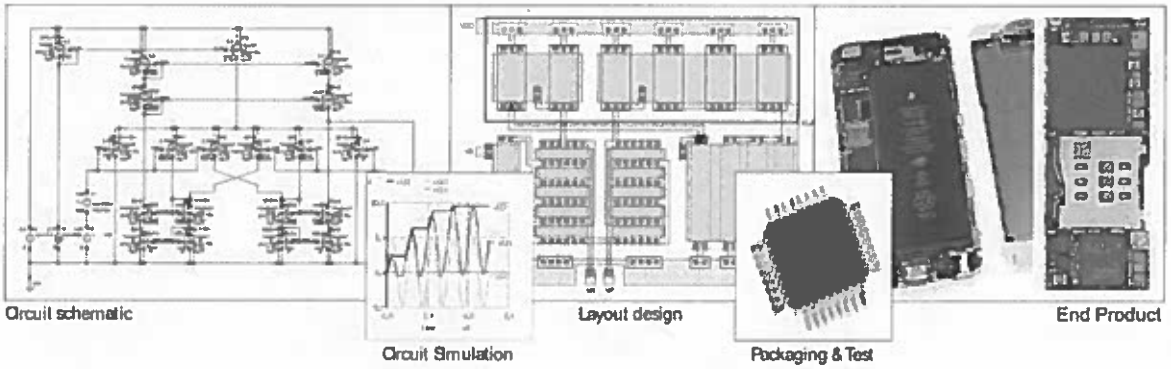


Fig. 1: IC design process: From design (schematic capture, simulation, layout), fabrication, packaging, test, etc, eventually to the end-product.

DACs are widely used in mixed-signal design to interface the digital and analogue domains. For example, all mobile phones use DACs to convert the digital input from a codec into a voltage output to drive the earphone. Different architectures and circuit designs for typical DACs are covered in the lectures E4.20/E9AC3, many of which are feasible for this project. However the DAC topology you implement for this project has to be one of the following 3 options:

1. R-2R Ladder Based DAC
2. Capacitor array based DAC
3. Current steering based DAC

The design of the internal blocks and choice of various design parameters are completely flexible and left entirely to you to design, as long as the design meets the technical specifications given in the following section.

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This coursework will count for **75%** of the total E4.20/E9AC3 module credit. You are required to **independently** design a complete DAC with serial input. You will need to define the system architecture, design and simulate the various circuit blocks to ensure it meets the required specifications and finally complete the full custom layout in the 0.18 μm process that you have already used in the laboratory.

2 Design Requirements

2.1 Overview and Input/Output Requirements

There are two sub-systems in this project, a digital interface to receive the serial data and generate a reference clock for synchronisation, and a DAC initiated by an external signal to convert the digital input signal to an analogue output. The correspondence between input digital code and analogue output voltage is shown in Fig.2. The output is given by:

$$VOUT = CODE[7:0]/255 + 0.4 \tag{1}$$

where CODE is the input vector and the relative output error for a given output is:

$$ERR_{rel} = abs(VOUT_{IDEAL} - VOUT_{OUT})/VOUT_{IDEAL} \tag{2}$$

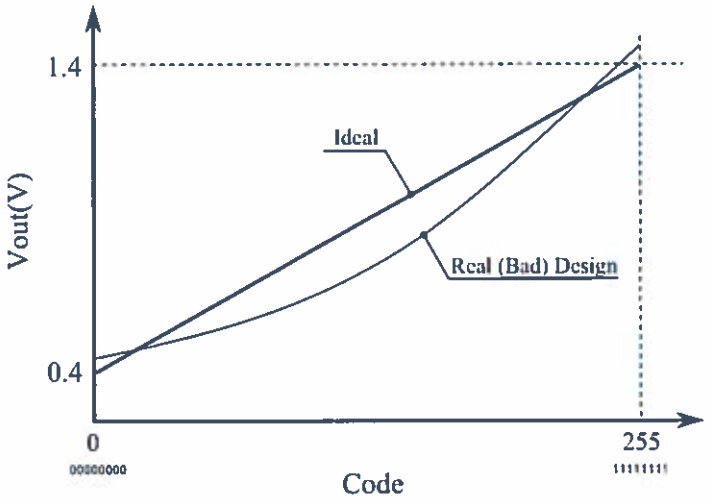


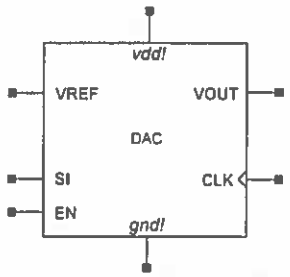
Fig. 2: Input/output characteristics for the DAC

The top level symbol of the DAC and pin description are shown in Fig. 3. Please note that in the top level symbol it is not necessary to place *vdd! gnd!* pins, as they will be by default connected to the top level (global) *vdd!* and *gnd!* nodes in the testbench.

2.2 Technical Specifications

The required technical specifications for the DAC are given in Table. 1, with the timing requirement shown in Fig. 4. The *EN* signal is active high for eight clock cycles and used to indicate that the data on *SI* is valid, and will be clocked in at each rising edge of CLK, with LSB first. During this time, VOUT can vary or remain stable. The time between the falling edge of EN and VOUT becoming valid (for a new input) has to be smaller than 1 μs. The absolute error

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Name	Type	Direction	Description
vdd!	Power	I/O	Global positive voltage supply – 1.8 V
gnd!	Power	I/O	Global negative/substrate voltage supply – 0 V
SI	Digital	In	Digital input, LSB leading
VREF	Analogue	In	Voltage reference at 0.9V
EN	Digital	In	High indicates the data on SI will be ready
CLK	Digital	Out	Clock signal at 10 MHz
VOUT	Analogue	Out	Voltage output corresponds to the input

Fig. 3: Top level symbol and pin descriptions for the system.

is also examined at this point. Please note that the analogue output should be available and stable before the next rising edge of EN even at fastest conversion rate, i.e 500 kHz, this 1 μs interval should be sufficient to establish the required conversion rate. It should be noted that

Table 1: Target technical specifications

Name	Specifications	Value	Units
Area	Total silicon area	$\leqslant$1	mm^2
VDD	Power Supply	1.8	V
P_{diss}	Power dissipation	$\leqslant$20	mW
f_{clk}	Output clock frequency,	10	MHz
f_s	Conversion rate	500	kHz
R	Output voltage range	0.4-1.4	V
Z_L	Load impedance for VOUT.	10k // 10	Ohm// pF
ERR_{rel}	Relative error	$\leqslant$1.5	%
T	Temperature	27	^oC

Logic 1 is 1.8 V; Logic 0 is 0 V
Analogue results will be tested 1 μs after the falling edge of EN.
See Fig. 4 for detail.

the load impedance for VOUT is external components and should not be included into your design.

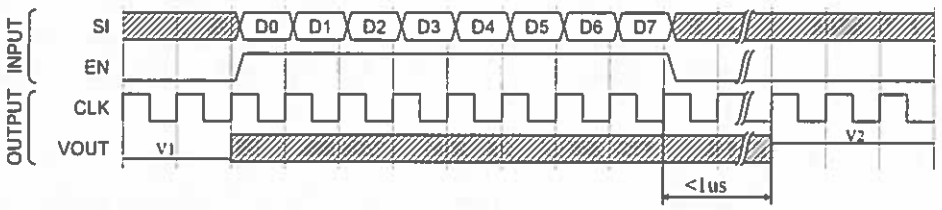


Fig. 4: Timing diagram for DAC. SI becomes valid on the next rising edge after EN is asserted high.

3 Guidelines

3.1 Design file organization

Creat the design library with name CMOS16_DAC_yourname, (replace yourname with your log in name, e.g. tc16). After creating the design library, make sure you place the provided top-level design cell (DAC) and testbenches (TB_DAC, test_DAC) in the design library folder, as mentioned in the appendix. The files and folders should look like as follows:

- Folder you run cadence

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- CMOS16_DAC_yourname ⇒ Design library
  - DAC
  - TB_DAC
  - test_DAC
  - ...
- Lab1 ⇒ Lab Libraries, do not refer cells in these from the design library
- ...

3.2 A good design

In general, circuit design is never perfect and always involves a number of trade-offs with no single, correct solution. For purposes of this coursework the following are expected (i.e. to achieve a good mark):

- Ensure your design does the job required. First and foremost, make sure you achieve what is asked before adding any extra functionality.
- Minimise power and area requirements. Less power consumption means an increased battery lifetime and less silicon area means lower cost.
- Messy designs very rarely work, in particular when systems become more complicated. Keep your schematics and layouts tidy, so that they can be reviewed easily.
- Clear, concise documentation can help others understand/follow your design as well as help you revise/reuse design in the future. It is essential you keep clear, consistent records.

3.3 System Level

Although you are required to choose the structure of your DAC from three options given previously, there are a number of different possible implementations for each structure with key design considerations that need to be made to meet the specification. You are encouraged to go through lecture notes and textbooks [2,3,5], or [7,8] for a general introduction (if textbooks are unavailable from the library).

It is highly recommended that before you dive into the schematic design, you first plan the high-level design architecturally (i.e. with pen and paper). This will help you define your I/O's, internal data/control lines, signal types, etc, and to specify the detailed technical specification for each block. You should spend some time reading relevant literature (i.e. text books, course notes, publications) to review possible circuit topologies and ensure your design is realistic and can meet the top level specifications. Please bare in mind that many of the specifications listed in Table. 1 can significantly impact your design. For example:

- The area will constrain the selection of both active and passive devices (i.e. component values and count), for example, a 0.8mm × 0.8mm capacitor is typically <1 nF.
- Power will limit the bandwidth of analogue components, which in turn limits the speed of conversion.

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3.4 Block level

Based on your selected topology, the sub blocks can be designed and configured in an infinite number of ways to achieve the given specifications. Always think how to **Reuse Your Design** in previous labs or in **Your** other blocks.

Several approaches for implementing the main blocks that may be required are listed as follows:

- Amplifier: This could use one of the following topologies: single transistor, operational amplifier, operational transconductance amplifier, transimpedance amplifier, etc. Each of these have different input/output impedance characteristics. Common structures include symmetric OTA, telescopic, two stage miller, folded cascode and etc, see ref [2,3] and lecture notes for details. Keep in mind that every type has its advantages and disadvantages, so you need to establish which suits your system. You have already designed a symmetric OTA in Lab 3 and can reuse this with any necessary modifications (if needed), or can alternatively implement an entirely new design according to the specifications.
- Clock: In Lab 2, you designed a 5-stage ring oscillator. This may need some modification/optimization before it can be included in this design. A precise 10 MHz clock is required within the specific area (in addition to the DAC core). You can either increase the delay between each stage or use an extra counter to do a frequency division.
- Digital blocks: All the required digital blocks, along with logic gates can be implemented based on standard topologies which are described extensively in most digital textbooks. It is highly recommended to use the digital cells that you have already designed in Labs 1-2. A set of basic core digital cells are provided with schematic and layout view, however, these designs are not perfect and you will receive a **5% penalty** if used (it is preferred if you instead use your own standard cell library).

3.5 Testbench

Testbenches are required for both system level and block level designs, although they will not be marked. For the system level, a ready-to-use test bench is provided, where all the pins correspond to the specification that has been given previously. In this project, you should design you own test benches to verify the functionality of each of the sub blocks. To generate a certain digital waveform, piecewise-linear voltage source vpwl in the analog lib and vbits can be used to generate certain waveforms, which can be linked to external data files.

4 Special Considerations

4.1 No Minimum Size for Analogue Circuits

In the given technology, the minimum transistor length is 0.18 μm, so you may wonder why not use this to achieve a high transconductance and also save area. However, as you know it is not possible to use minimum size transistors for analogue design due to mismatch. As you may recall, there is an area dependance on mismatch, i.e. matching between two transistors is inversely proportional to the square root of the area. Although you are not required to make mismatch calculations in your design process, you should use devices that are at least 3 times

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the minimum size (both in width and length) to ensure good matching (for all components including transistors, capacitors and resistors).

Similarly, for analogue circuits, it is recommended to avoid using minimum width metal tracks and not to route over active devices (to minimise parasitic resistances/capacitances). Although you are not required to extract parasitics, you are recommended to use a width of 2× the minimum for signal, and 5× for power tracks.

4.2 Non-linearity

Non-linearity is one of the major contributors of distortion in data converters (with the other being noise). For purposes of this coursework, only the maximum absolute relative error is tested. Usually this occurs when the MSB changes. However, the amplifier can also introduce non-linearity if the open loop gain is small.

4.3 Speed

Given the required sampling frequency, the data conversion time should be smaller than the sampling period. This limits the digital blocks, converter and amplifier speed. Increasing power is the most straightforward way to increase the speed, or by reducing any passive component values (if used) and/or increasing transistors W/L can help as well.

4.4 Drive Strength

Drive strength is a major concern if you have various load conditions in your analogue circuits. Extremely small or large impedances can affect your amplifier gain and bandwidth, and also corrupt any feedback. Bare in mind that a real amplifier does not exhibit ideal characteristics (eg. infinite gain, infinite input impedance, zero output impedance, etc). In order to achieve the required drive strength, the maximum source/sink current of the output stage and amplifier topology needs to be considered.

4.5 Layout

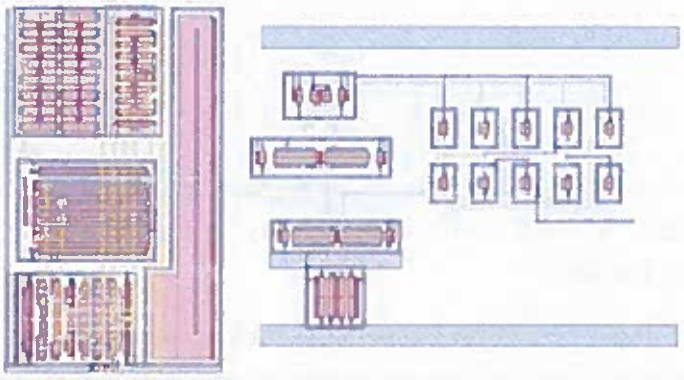
Analogue layout is a combination of good engineering and creativity (art!). Generally, a compact and organised layout will result in better circuit performance and a reduced total area, which means a more efficient implementation. Even for analogue layout, a standardised layout scheme is extremely beneficial. This may include imposing a consistent block height, routing scheme (different metal for horizontal and vertical routing), and I/O positions, etc. It will save you a lot of time in the top level layout if all your sub-blocks conform to the same scheme.

It is expected to apply layout techniques used in the lab session, e.g common-centroid and dummy devices, such that the yield of devices with required dimensions can be improved. It is also important to mention that the blank area in your layout is actually silicon substrate or well. The potential (i.e. voltage) of this silicon area can affect the performance of the transistors, therefore it is critical to add as many substrate or well contacts as possible to maintain the potential and sink any spontaneous current. These fixed-bias contacts are also helpful in minimising the risk of latch up.

A comparison of good and bad layouts can be seen in Fig. 5. It can be observed that the silicon area is used more efficiently, with a lot of contacts to maintain a consistent potential across the well and substrate. Also the transistors are in a similar orientation for P and N

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MOSFETs respectively, with clearly organised metal tracks connecting all the devices.



a) b)

Fig. 5: Comparison between: (a) good; and (b) bad layout.

4.6 Design Files

Ensure your designs for each technology are kept in a different folder. Never mix process technologies. For this coursework organize your design files/cells in a single library, with all relevant test benches. For this project do NOT create multiple libraries, except the CORELIB_GPDK18 provided. If you used the cells designed in the lab session, please copy the relevant designs and related hireacy files into your project design library. Back up your design when necessary, this can be done by zipping the entire library, or creating new a backup cellview for each cell.

5 Administration and Deliverables

Each student has to independently design the DAC including the architecture, schematics, simulations, and layout. If you organise your time well, it is estimated you will need 60 hours work to complete this coursework. As in industry, you will have milestones as listed in Table. 2, with three deliverables: preliminary design review, design files and the final report.

Table 2: Deadlines and Assessed Deliverables

Week	Data	Deliverable
5	11 Nov. 2016 @ 21:00	Interim Design Review Presentation (via Blackboard) Oral times TBC (to be scheduled week starting 14 Nov.)
9	2 Dec. 2016 @ 21:00	Design Submission (via Blackboard)
11	16 Dec. 2016 @ 21:00	Final Report (via Blackboard)

5.1 Preliminary Design Review

At the first milestone you need to give a presentation to explain your design. You will be required to submit (via Blackboard) a maximum of 5 slides (with no animations) by the deadline, and the date/time of your oral presentation will be scheduled soon after. The oral test will consist of a 5 minute presentation followed by 10 minutes of questions.

In your slides it is recommended you include the following items:

- Top level (internal) architecture showing I/O ports and with corresponding timing diagram;

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- Annotated circuit schematics (in Cadence);
- Preliminary simulation results;
- Detailed work plan showing where you are so far and what is left to complete

The Preliminary Design Review will carry a 20% weighting of the entire course. The marking scheme is listed in Tab.3. You will receive a **5% penalty** if the presentation exceeds five minutes or you use more than five slides.

Table 3: Marking scheme for preliminary design review

Content	Percent
Quality and design innovation	40%
Feasibility - will it work, and will it be complete in time constraints?	30%
Engineering competence and understanding of full custom IC design	30%

5.2 Design Submission

Design submission includes (via Blackboard):

- A .tar.gz file containing your complete design library. This can be done in your cadence run directory (where you invoke ic6) by:

  tar -cvf yournameCID.tar ./yourlibraryname

  You will see a list of files and folders are packed into a single file, then

  gzip yournameCID.tar

- A design brief document detailing the structure of the library path, i.e which is the top level system, which is the op-amp, etc, also including all the main schematics (2 page maximum). This document will be used as reference for your design and will NOT be marked.

It should be noted that you need to run the simulation script as described in the Appendix, such that design files will be checked and simulation results in text files will be included into your library folder.
The Design Evaluation will carry a 25% weighting on the entire course, the marking scheme is listed in Tab.4. You will receive a **5% penalty** if you use the standard cell library that is provided.

Table 4: Marking scheme for design submission

Content	Percent
Design and functionality - innovation, does it work? Standard cells used?	25%
Specifications - have design targets been met (or beaten)?	25%
Design clarity - good symbols, well annotated schematics - easy to follow	25%
Layout - good practice, DRC/LVS clean	25%

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5.3 Final Report

At the final milestone you need to submit the Final Report (in PDF format): This must be compliant with the IEEE journal style (template available from IEEE in both MS word and LaTeX formats) with a page limit of 4 pages (see example in Sec.5.3). A suggested report *structure* is as follows: Abstract, (I) Introduction, (II) System Overview, (III) Circuit Implementation, (IV) Simulated Results, (V) Conclusion. The report should include the following *content*:

- Conceptual block diagram indicating how your design works.
- Annotated schematics and layout.
- Simulation results to demonstrate that the circuit operates as expected.
- Description of the test strategy used and a schematic test bench.
- Discussion of advanced issues considered/addressed and unique features and limitations of your design.
- Table of technical specifications.

The Final Report will carry a 30% weighting on the coursework assessment, the marking scheme is listed in Tab.5.

Table 5: Marking scheme for design report

Content	Percent
Design description	25%
Circuit theory of operation	25%
Results and Interpretation/discussion	25%
Presentation	25%

References

[1] MJM Pelgrom, "Analog-to-Digital Conversion" *Springer*, 2010.

[2] PE Allen and DR Holberg, "CMOS analog circuit design" *Oxford University Press (2nd ed.)*, 2002.

[3] RJ Baker, "CMOS: Circuit design, layout, and simulation" *Wiley-IEEE Press (3rd ed.)*, 2011.

[4] A Hastings, "The art of analog layout" *Prentice Hall (2nd ed.)*, 2006.

[5] RJ Van de Plassche "CMOS integrated analog-to-digital and digital-to-analog converters" *Kluwer Academic Publishers Dordrecht (2nd ed.)*, 2003.

[6] "http://www.maximintegrated.com/app-notes/index.mvp/id/283".

[7] "http://www.aicdesign.org/SCNOTES/2006notes/Chap10(12_8_06).pdf".

[8] "http://wiki.analog.com/university/courses/tutorials/cmos-dac-chapter".

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How to include the standard cell library

The provided standard cell library includes essential digital building blocks, such as logic gates and flip-flops. To use this library, you firstly need to extract the zip file under the **directory where you run cadence**, then include this folder into your library as shown in Fig.6. You can add the instance such as INV8 for inverter shown in Fig.7, into your own schematics, remember to run simulation to make sure the cell you choose will be suitable for your system.

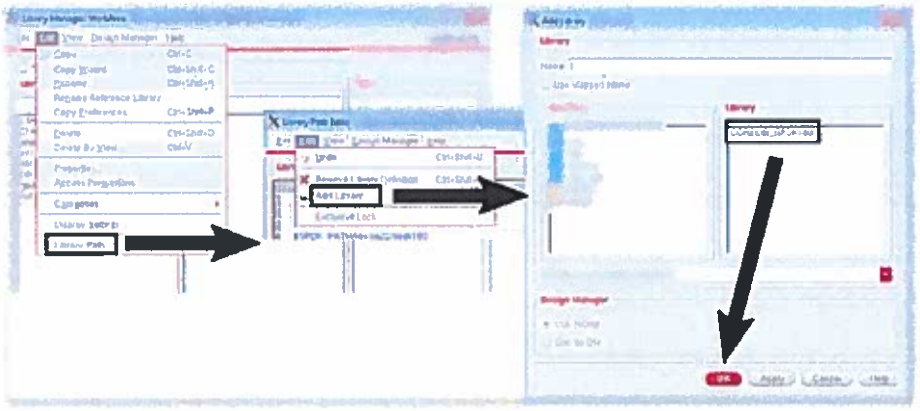


Fig. 6: Load library path in the Library manager

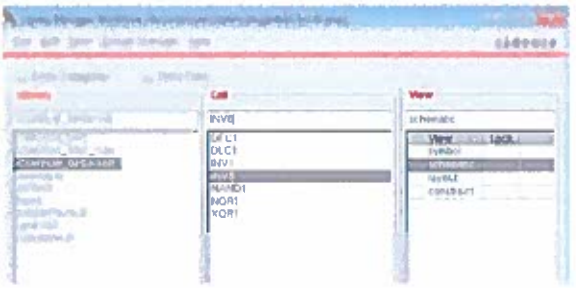


Fig. 7: Put TB_DAC under your project library

How to use the testbench

The attached testbench is a *veriloga* model, which generates the necessary digital IO signal, EN, SI, which is synchronized to the clock input. It also generates a reference output corresponding to the digital signal SI in each sampling period. You task is to make sure your DAC output as closely as possible match this reference output.

To use this test bench, you firstly need to extract the zip file under your **library folder**, make sure TB_DAC is directly under your Library folder (e.g. CMOS16_DAC_tc16). You should able to check this with library manager, see Fig.8. You can then include the symbol in your test circuits with correct pin connections. To simulate the schematic with the *veriloga* model, you need to include the *veriloga* view into the view list, ADE, see Fig. 9. After that the simulation can be run in the way same as in the labs.

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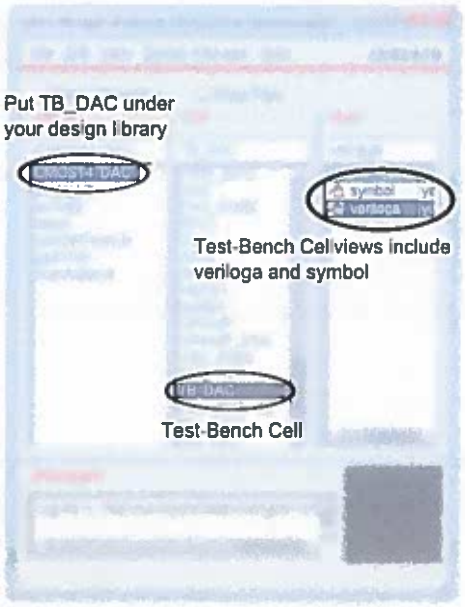


Fig. 8: Put TB_DAC under your project library



Fig. 9: Set up the simulation with *veriloga* models included

How to use the test script to generate the simulation report

A *SKILL* based simulation script is provided along with the test bench circuits. It creates a new schematics with provided test bench circuits, copies the symbol view of your design into it, connect all the wires according the port names, runs the DC and transient simulation and generates simulation report in a text file.

To use this script, please follow these steps:

1. Copy the script file *.il into the directory you run Cadence from.
2. Make sure you have test bench files and design files ready as explained in the previous section.
3. Load the script by typing:

   load("testDAC.il")

   in the CIW window as in Fig. 10
4. Run the script:

   YLCTestDAC("designlib" "topdaccell" "yourname")

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in the CIW window, with *designlib* is your design library, *topdaccell* is the top design cell including a symbol as shown in Fig. 3, and *yourname* is your login name for using the PC, for example *YLCTestDAC("CMOS16_DAC_tc16" "DAC_tc" "tc16")*

5. The generated results files are under ./libname/results, please do NOT change these files.



Fig. 10: Load the script then run by typing the command in the CIW window

If your design successfully simulates, you can find the test results, and text will be displayed to indicate your simulation has completed with all cell views correctly connected. This can help you evaluate your design files before submission. A list of common error are as follows:

1. Cellview of schematic in design lib can not be found or opened: check if your design cell is in the correct library, with symbol view and corresponding schematic view.
2. Unmatched to any pin: Pin names is not labelled as in Fig. 3, re-label as stated in the table.
3. Clock start failed: Simulation stops due to oscillator fails, the start-up condition of your oscillator may not be correctly designed.

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Example Final Report (from Previous Years Fig. 11)

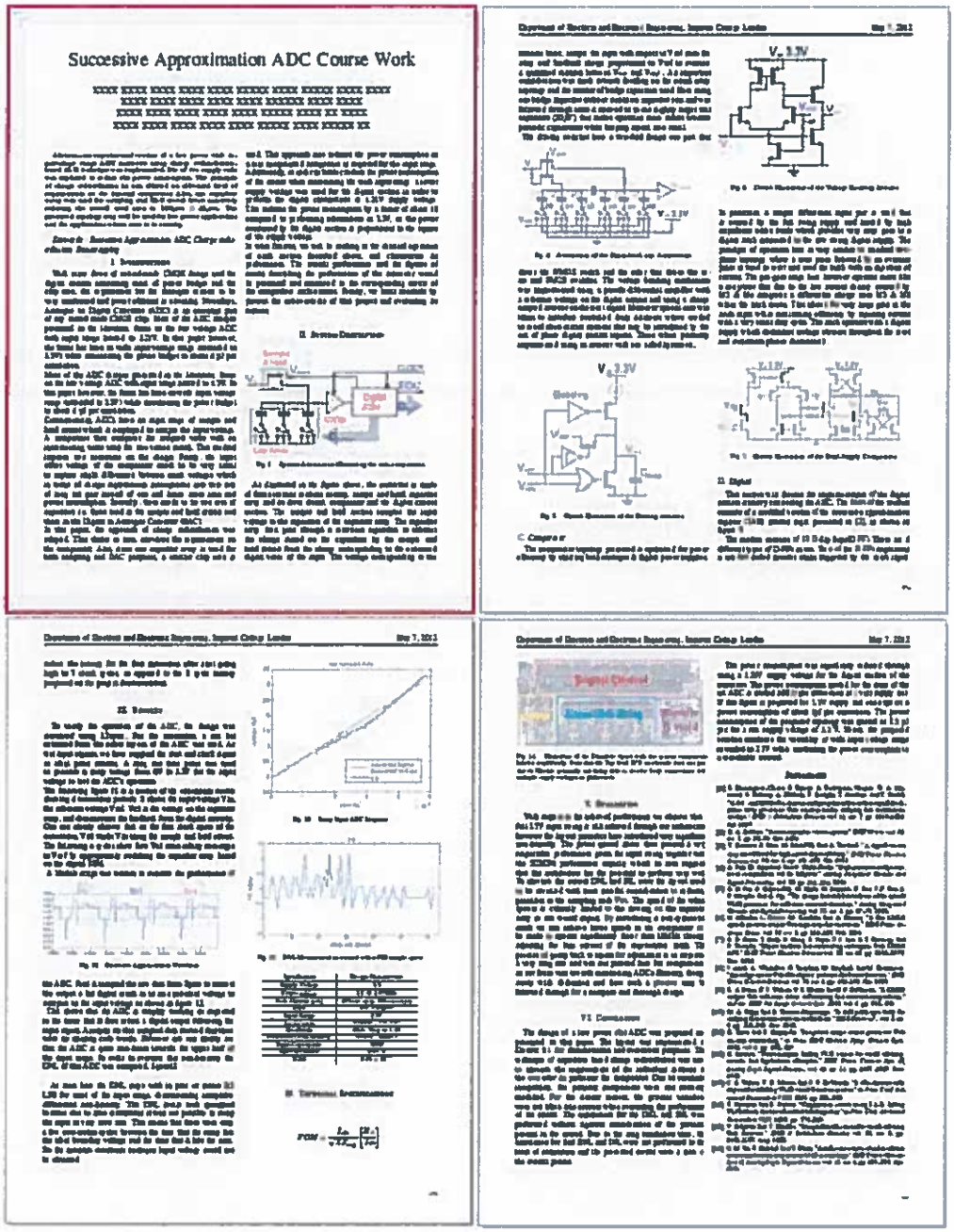


Fig. 11: Final report example